

Krishnaa Vadivel | Semiconductor Physics | Integrated Photonics | Thin-Film Fabrication

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Summary

Electrical engineering PhD researcher with Cornell and Yale ECE training across thin-film fabrication, semiconductor physics, integrated photonics, device simulation, and scientific computing. Experience spans cleanroom GaN device development, optical and materials characterization, light-matter modeling, and distributed HPC software for semiconductor and photonic systems.

Experience

Yale University, Electrical Engineering

PhD Research Assistant

2021–present

- Developed GaN-based power-electronics devices in cleanroom and MOCVD environments during the early PhD, including selective-area etching techniques and materials characterization using SEM, cathodoluminescence, and Raman spectroscopy.
- Modeled VCSEL emission and related light-matter interactions involving coupled photon, phonon, and electron dynamics for semiconductor and photonics problems.
- Built distributed HPC software for exciton-polaron and light-matter interaction modeling using first-principles methods, quantum many-body theory, and accelerator-enabled scientific computing.
- Used agentic ML tooling to generate first-principles software inputs from journal literature and technical documentation.
- Co-authored a 2025 *Journal of the American Chemical Society* paper and delivered APS presentations in 2024, 2025, and 2026 on self-trapped exciton formation and related excited-state materials physics.

Probe Technologies Inc. / Weatherford Wireline Services

Software and Embedded Systems Engineer

2019–2021

- Parallelized CUDA software for analyzing and visualizing acoustic and nuclear well-tool data used in engineering diagnostics and field interpretation.
- Applied dolfinx-based finite-element analysis to frequency studies of steel pipes in well environments.
- Supported instrumentation-oriented engineering around deployed well tools, connecting field data, sensor systems, and numerical analysis.

FindLight

Photonics Technical Writer

2018–2019

- Wrote technical content on lasers, integrated optics, and optoelectronic components for engineering-facing audiences.

Selected Projects

Integrated Photonics Band-Pass Filter: Designed a Meep-based integrated photonics band-pass filter, then fabricated and tested the device with an erbium-doped fiber laser setup.

Semiconductor Simulation: Developed numerical modeling of semiconductor electrostatics, drift-diffusion transport, and pn-junction-style device behavior.

DOLFINx Finite-Element Analysis: Built 2D DOLFINx/PETSc simulations for transient heat diffusion, Poisson potential around a circular cutout, and incompressible flow past an immersed cylinder, with PyVista and Matplotlib results.

VCSEL Light-Matter Modeling: Built simulation tools for VCSEL emission and coupled photon, phonon, and electron interactions in semiconductor systems.

Equivariant Scientific ML: Developed PyTorch Geometric equivariant neural networks for material energy prediction and optical-property prediction.

Education

Yale University

PhD, Electrical Engineering

2021–present

Yale University

MPhil, Electrical Engineering

2023

Yale University

MS, Electrical Engineering

2023

Cornell University

MEng, Electrical and Computer Engineering

2017–2018

Cornell University

BS, Electrical and Computer Engineering

2013–2017

Technical Skills

Fabrication / Characterization: Cleanroom processing, MOCVD, SEM, cathodoluminescence, Raman spectroscopy, laser-bench optical testing

Devices / Photonics: Semiconductor physics, thin films, integrated photonics, lasers, VCSEL modeling, light-matter interaction

Simulation: Semiconductor device simulation, FDTD, finite-element analysis, first-principles and quantum many-body modeling

Machine Learning: PyTorch, PyTorch Geometric, equivariant graph neural networks, optical- and materials-property prediction

Software / HPC: Python, C, C++, CUDA, Linux, MPI, OpenMP, PETSc, SLEPc, Docker